Why Do AI Language Models Hallucinate?

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Problem & Argument

- Many Large-Language Models (LLMs) hallucinate; provide plausible falsehoods
- Statistical pressures encourage hallucinations
  - Pre-training learns distribution of language in text; certain statements have little patterns between correct & incorrect
  - Tend to predict plausible statements even if content is incorrect
- Metrics optimized during post-training necessarily encourage hallucinations
  - Guessing even when unsure maximizes reward under binary metrics (e.g. accuracy)
  - Most primary evaluations used penalize uncertainty

Pre-training (w/out prompts)

- Pre-trained language model determines probability distribution over text
    - Partition text into $\mathcal{X} = E \cup V$; E as errors, V as valid examples; error rate as $p(E)$.
    - Consider training data as noiseless distribution $p(\mathcal{X})$ wherein $p(E) = 0$
- Consider Is-It-Valid (IIV) classification problem
    - Probability distribution over examples $\mathcal{X}$, as 50/50 mix of samples from p and uniformly random errors
    - $D(x) := \{ p(x)/2$ if $x \in V$, $1/(2|E|)$ if $x \in E$
    - Target function
    - $f(x) := \{ +$ if $x \in V$, $-$ if $x \in E$

Pre-training (w/out prompts)

- Use language model as IIV classifier
  - err_{iiv} = Pr[$\hat{f}$(x) != f(x)] wherein $\hat{f}$(x) := {+ if $\hat{p}$(x) > 1/|E|, - if $\hat{p}$(x) <= 1/|E|}
- Error rate $\hat{p}(E)$ is necessarily bounded by error in IIV problem
  - err >= 2*err_{iiv} - |V|/|E| - δ, wherein δ = |$\hat{p}(A)$ - p(A)| for A := {x ∈ $\mathcal{X}$ | $\hat{p}$(x) > 1/|E|}
- For pre-trained model wherein |$\hat{p}(A)$ - p(A)| is small, model will tend to err
- For unlearnable concepts wherein |V|/|E| is small and err_{iiv} is large, model will tend to err

Pre-training (w/ prompts)

- Contexts $c \in C$ drawn from prompt distribution μ; each x consists of (c, r) prompt & plausible response
  - For given c, $V_c := \{r \mid (c, r) \in V\}$ as valid responses $E_c := \{r \mid (c, r) \in E\}$ as errors
  - $p(r \mid c)$, $\hat{p}(r \mid c)$ as conditional probability distributions
  - $err := \Sigma_{(c,r)\in E}\,\mu(c)\hat{p}(r \mid c)$
  - err_{iiv} same as before, wherein $f(c,r) := \{+ \text{ if } \hat{p}(r \mid c) > 1/\min_c|E_c|, - \text{ if } \hat{p}(r \mid c) <= 1/\min_c|E_c|\}$
  - $D(c, r) := \{p(c, r)/2 \text{ if } (c, r) \in V,\ \ \mu(c)/(2|E_c|) \text{ if } (c, r) \in E\}$
  - $err >= 2{*}err_{iiv} - \max_c|V_c|/\min_c|E_c| - \delta$, wherein $\delta := |\hat{p}(A) - p(A)|$ for $A := \{(c, r) \in \mathcal{X} \mid \hat{p}(r \mid c) > 1/\min_c|E_c|\}$

Error Factors

- Extremely rare & specific arbitrary facts
  - E.g. birthdays of lesser-known individuals
- Model poorly fitted to specific concept
  - Counting letters when LLMs divide text into "tokens"
- Computational Hardness
- Distribution Shift (unseen testing examples)
- Replicating errors in training data (noisy datasets IRL)

Post-training

- Binary metrics penalize IDK responses, making overconfident guesses optimal
    - Given prompt c, plausible responses as $R_c := \{r \mid (c,r) \in \mathcal{X}\}$, abstention responses as $A_c \in R_c$
    - A grader $g_c(r)$ is binary if $\{g_c(r) \mid r \in R_c\} = \{0,1\}$ and $g_c(r \in A_c) = 0$
    - Model's beliefs as posterior distribution p_c over binary graders
    - Optimal responses are not abstentions $$\mathcal{A}_c \cap \arg\max_{r \in \mathcal{R}_c} \mathop{\mathbb{E}}_{g_c \sim \rho_c} [g_c(r)] = \emptyset$$
- Can integrate explicit confidence targets
    - Penalize mistakes as t / (1-t) points w/ t as target confidence/probability threshold
    - Mention explicitly in instructions during post-training

Experiments

Prompt: Adam Tauman Kalai's Ph.D. dissertation (completed in 2002 at CMU) is entitled: (do not search the web)

- GPT-5.3: "Efficient Learning in Large-Scale Systems."
- DeepSeek: "Efficient Algorithms for Learning and Optimization."
- Correct answer:
  - Probabilistic and On-line Methods in Machine Learning

Results & Key Takeaways

- Pre-training necessarily introduces errors/hallucinations in generation
- Current metrics for post-training encourage guessing over abstention
- Need to adjust post-training metrics to allow abstention and penalize guessing
- Incorporate is-it-valid performance into model evaluations